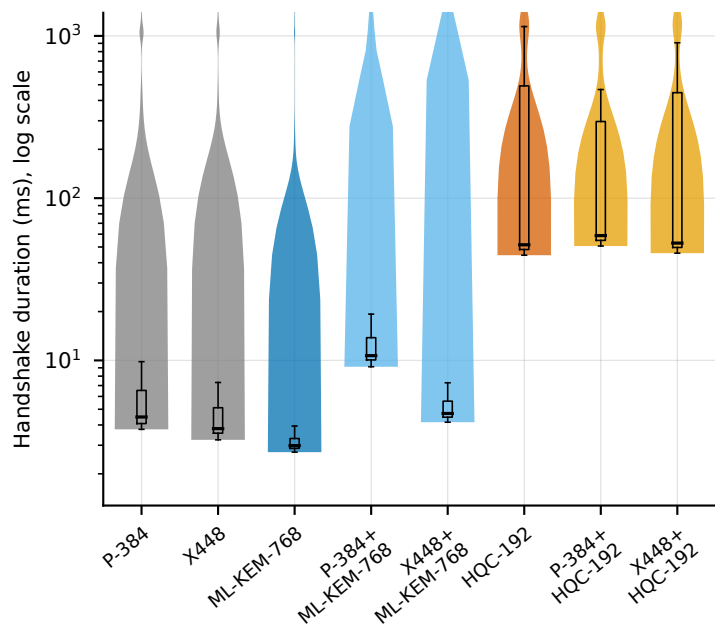
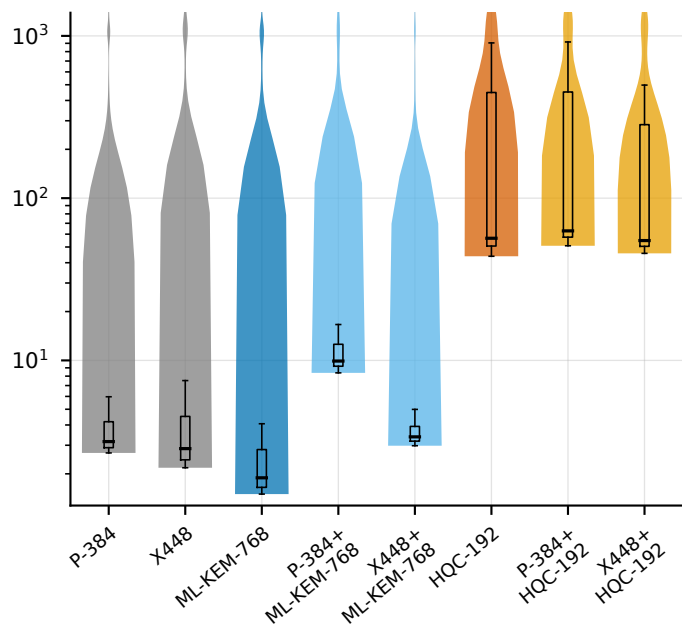


**(a)**

secp384r1 (classical)

**(b)**

mlDSA65 (post-quantum)

**(c)**
 Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM              | secp384r1<br>Mean (ms) | secp384r1<br>CV (%) | secp384r1<br>Max (ms) | mlDSA65<br>Mean (ms) | mlDSA65<br>CV (%) | mlDSA65<br>Max (ms) |
|------------------|------------------------|---------------------|-----------------------|----------------------|-------------------|---------------------|
| P-384            | 131.9                  | 264.8               | 3.30k                 | 141.7                | 295.3             | 3.74k               |
| X448             | 131.7                  | 314.8               | 3.30k                 | 203.8                | 275.6             | 7.82k               |
| ML-KEM-768       | 55.7                   | 437.9               | 2.08k                 | 178.3                | 312.1             | 7.64k               |
| P-384+ML-KEM-768 | 232.9                  | 564.6               | 26.55k                | 195.2                | 366.5             | 11.44k              |
| X448+ML-KEM-768  | 226.7                  | 1050.1              | 52.41k                | 95.9                 | 439.4             | 6.60k               |
| HQC-192          | 439.0                  | 171.1               | 5.46k                 | 498.1                | 240.5             | 14.65k              |
| P-384+HQC-192    | 368.1                  | 159.6               | 4.64k                 | 458.4                | 205.6             | 13.11k              |
| X448+HQC-192     | 398.4                  | 164.2               | 4.50k                 | 406.3                | 190.6             | 6.59k               |